CLOUD SECURITY PBL

UNIT 2 — Evidence & Explanation Report

IAM & Least Privilege Enforcement

Student Details

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Included Evidence

- PBL-Admins, PBL-Developers and PBL-Auditors groups
- Administrator group-based permissions
- Developer and Auditor least-privilege policies
- EC2-to-S3 IAM role and instance profile
- Admin MFA device assignment evidence

Submission Status

MFA device assignment for pbl-terraform-admin is now included.
If the instructor requires policy-level deny-without-MFA enforcement, that separate policy evidence is not shown in the supplied screenshots.

Prepared from the supplied project evidence. No university name included.

UNIT 2 — Evidence Summary

No.	Evidence	Purpose / Section	Status
1	Initial Admin Group Creation	Unit 2 → IAM Groups	Current evidence
2	IAM Groups — Admins, Auditors and Developers	Unit 2 → IAM Groups	Current evidence
3	IAM Users	Unit 2 → IAM Users	Current evidence
4	Terraform Administrator User	Unit 2 → Admin User / Group-Based Permissions	Current evidence
5	Auditor and Developer Least-Privilege Policies	Unit 2 → Least-Privilege Policies	Current evidence
6	Developer, EC2-S3 Policies and EC2 Role	Unit 2 → Custom Policies and EC2 Role	Current evidence
7	EC2 Role Policy Attachments	Unit 2 → EC2 IAM Role	Current evidence
8	EC2 Role Policy Attachments — Duplicate Evidence	Unit 2 → EC2 IAM Role	Duplicate/reference
9	Least-Privilege Policy Plan — Reference Copy	Unit 2 → Least-Privilege Policies	Duplicate/reference
10	Custom IAM Policy Plan — Reference Copy	Unit 2 → Custom Policies	Duplicate/reference

Evidence 1: Initial Admin Group Creation

Place in: Unit 2 → IAM Groups

Status: Current evidence

What it proves: Shows creation of the PBL-Admins IAM group.

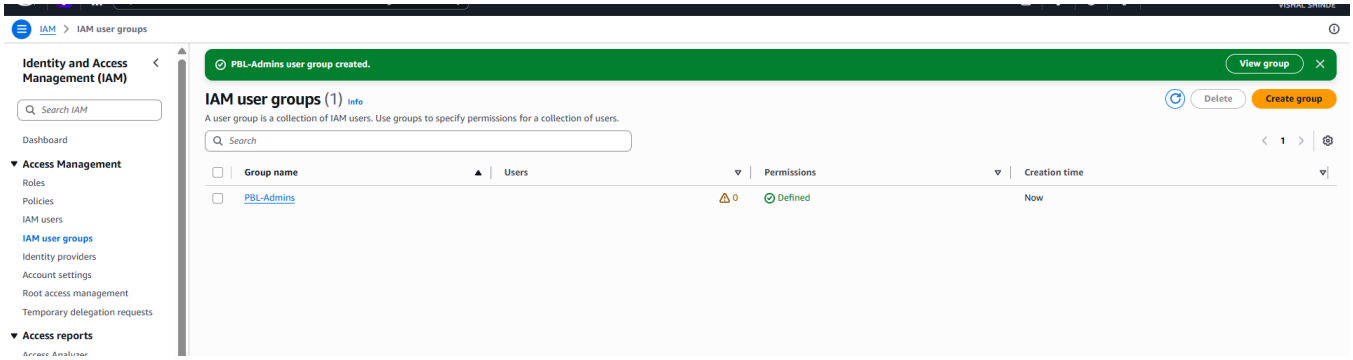


Figure 1: Initial Admin Group Creation.

Evidence 2: IAM Groups — Admins, Auditors and Developers

Place in: Unit 2 → IAM Groups

Status: Current evidence

What it proves: Shows all three required IAM groups: PBL-Admins, PBL-Auditors and PBL-Developers.

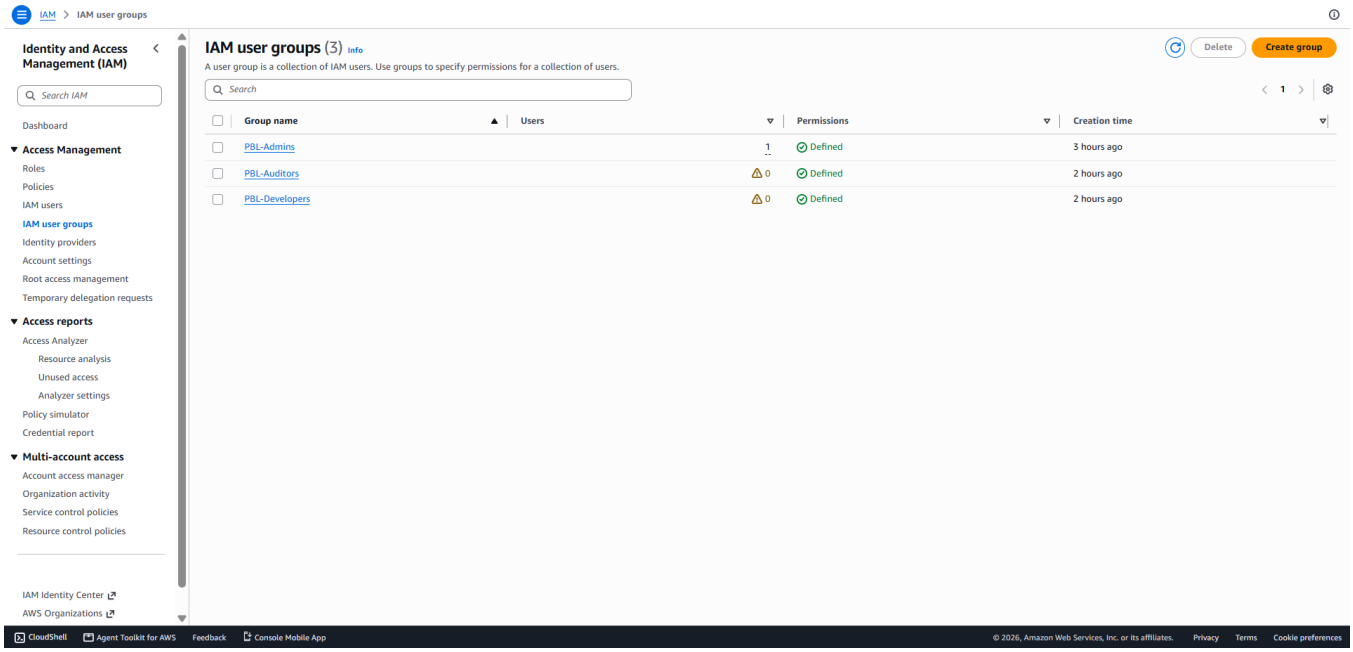


Figure 2: IAM Groups — Admins, Auditors and Developers.

Evidence 3: IAM Users

Place in: Unit 2 → IAM Users

Status: Current evidence

What it proves: Shows IAM users and the Groups column, providing evidence of IAM user administration.

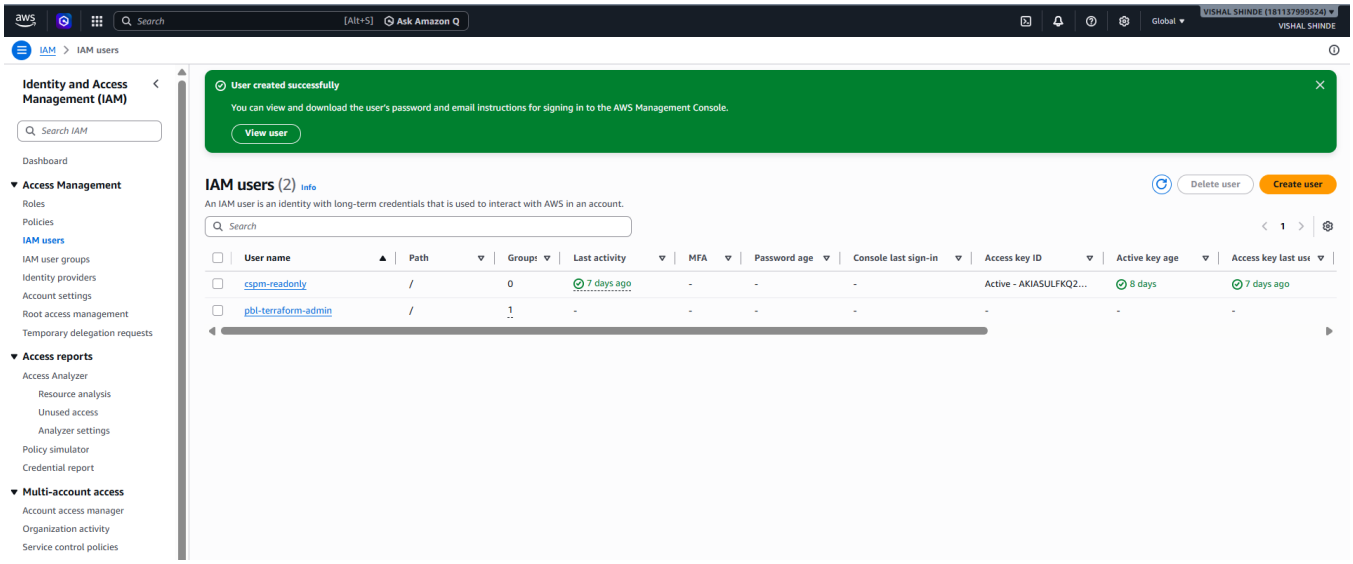


Figure 3: IAM Users.

Evidence 4: Terraform Administrator User

Place in: Unit 2 → Admin User / Group-Based Permissions

Status: Current evidence

What it proves: Shows the pbl-terraform-admin user and that AdministratorAccess is attached through the PBL-Admins group.

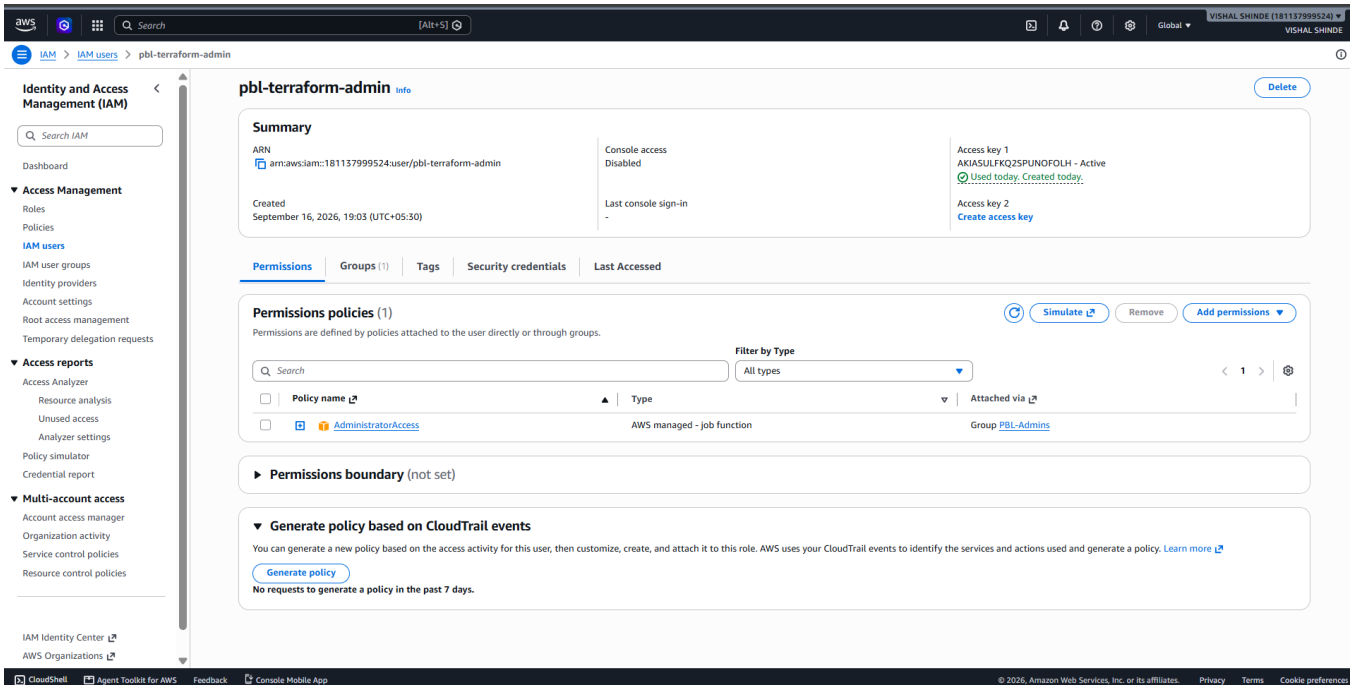


Figure 4: Terraform Administrator User.

Evidence 5: Auditor and Developer Least-Privilege Policies

Place in: Unit 2 → Least-Privilege Policies

Status: Current evidence

What it proves: Shows Terraform definitions for PBL-Auditor-ReadOnly and PBL-Developer-LeastPrivilege and their group attachments.

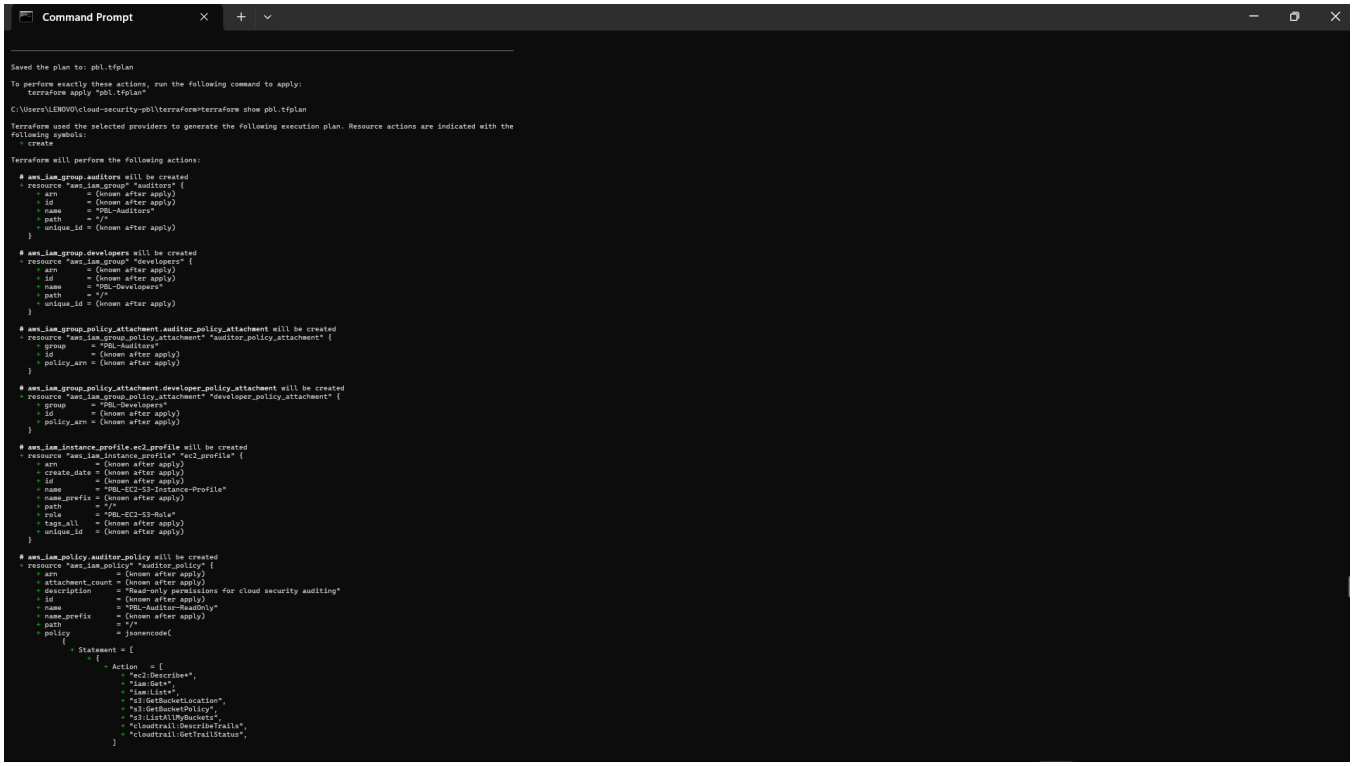


Figure 5: Auditor and Developer Least-Privilege Policies.

Evidence 6: Developer, EC2-S3 Policies and EC2 Role

Place in: Unit 2 → Custom Policies and EC2 Role

Status: Current evidence

What it proves: Shows the custom Developer policy, EC2-S3 least-privilege policy and PBL-EC2-S3-Role trust relationship.

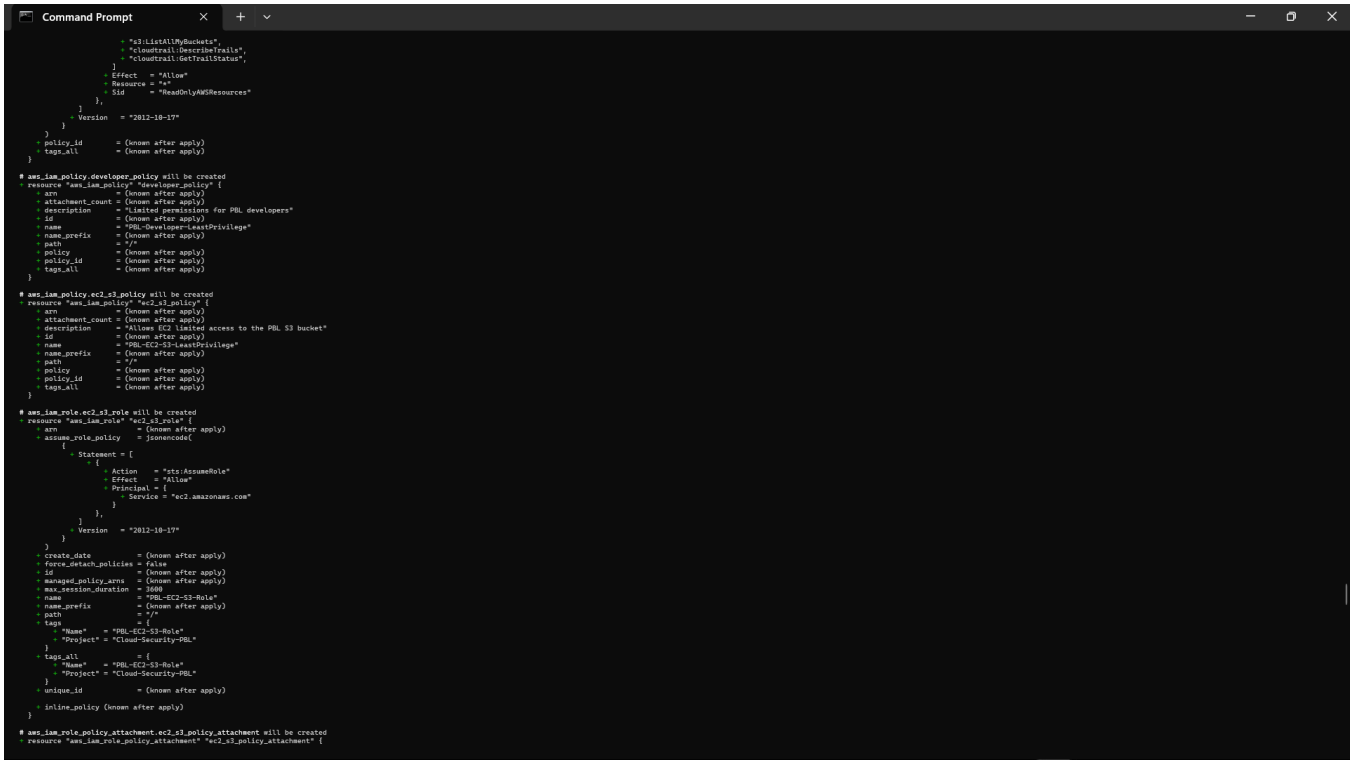


Figure 6: Developer, EC2-S3 Policies and EC2 Role.

Evidence 7: EC2 Role Policy Attachments

Place in: Unit 2 → EC2 IAM Role

Status: Current evidence

What it proves: Shows PBL-EC2-S3-LeastPrivilege and AmazonSSMManagedInstanceCore attached to the EC2 role.



```
# aws_iam_role_policy_attachment.ec2_s3_policy_attachment will be created
+ resource "aws_iam_role_policy_attachment" "ec2_s3_policy_attachment" {
    + id         = (known after apply)
    + policy_arn = (known after apply)
    + role       = "PBL-EC2-S3-Role"
  }

# aws_iam_role_policy_attachment.ec2_ssm_policy_attachment will be created
+ resource "aws_iam_role_policy_attachment" "ec2_ssm_policy_attachment" {
    + id         = (known after apply)
    + policy_arn = "arn:aws:iam::aws:policy/AmazonSSMManagedInstanceCore"
    + role       = "PBL-EC2-S3-Role"
  }

# aws_instance.pbl_web_server will be created
+ resource "aws_instance" "pbl_web_server" {
    + ami                                  = "ami-0efe89789aa766662"
    + arn                                  = (known after apply)
    + associate_public_ip_address          = true
    + availability_zone                    = (known after apply)
    + disable_api_stop                     = (known after apply)
    + disable_api_termination              = (known after apply)
    + ebs_optimized                        = (known after apply)
    + enable_primary_ipv6                  = (known after apply)
    + force_destroy                        = false
    + get_password_data                    = false
    + host_id                              = (known after apply)
    + host_resource_group_arn              = (known after apply)
    + iam_instance_profile                 = "PBL-EC2-S3-Instance-Profile"
    + id                                   = (known after apply)
    + instance_initiated_shutdown_behavior = (known after apply)
    + instance_lifecycle                   = (known after apply)
    + instance_state                       = (known after apply)
    + instance_type                        = "t3.micro"
    + ipv6_address_count                   = (known after apply)
    + ipv6_addresses                       = (known after apply)
    + key_name                             = (known after apply)
    + monitoring                           = true
    + outpost_arn                          = (known after apply)
    + password_data                        = (known after apply)
    + placement_group                      = (known after apply)
    + placement_group_id                   = (known after apply)
    + placement_partition_number           = (known after apply)
    + primary_network_interface_id         = (known after apply)
    + private_dns                          = (known after apply)
    + private_ip                           = (known after apply)
    + public_dns                           = (known after apply)
    + public_ip                            = (known after apply)
    + region                               = "ap-south-1"
    + secondary_private_ips                = (known after apply)
    + security_groups                      = (known after apply)
    + source_dest_check                    = true
    + spot_instance_request_id             = (known after apply)
    + subnet_id                            = (known after apply)
    + tags                                 = {
        + "Environment" = "Lab"
        + "Name"        = "PBL-Web-Observation-EC2"
        + "Project"     = "Cloud-Security-PBL"
        + "Purpose"     = "Web and SSH Threat Observation"
      }
    + tags_all                             = {
        + "Environment" = "Lab"
        + "Name"        = "PBL-Web-Observation-EC2"
        + "Project"     = "Cloud-Security-PBL"
        + "Purpose"     = "Web and SSH Threat Observation"
      }
    + tenancy                              = (known after apply)
    + user_data                            = <<-EOT
        #!/bin/bash

        dnf update -y

        dnf install -y nginx

        systemctl enable nginx
        systemctl start nginx

        cat > /usr/share/nginx/html/index.html <<'HTML'
        <!DOCTYPE html>
        <html>
        <head>
            <title>Cloud Security PBL</title>
        </head>
```

Figure 7: EC2 Role Policy Attachments.

UNIT 2 — Submission Notes

Unit 2 evidence covers IAM groups, users, group-based administrator access, least-privilege policies, EC2 IAM role/instance profile, and Admin MFA.

Admin MFA evidence:
pbl-terraform-admin has one passkey/security-key MFA device assigned, as shown in the AWS IAM Security Credentials screenshot on the next page.

Important accuracy note:
The supplied screenshot proves MFA device assignment. It does not, by itself, prove a policy-level DenyWithoutMFA enforcement rule. Add that policy screenshot only if the instructor explicitly requires policy-level enforcement.

Security note: do not publish AWS access-key material or other credentials.

UNIT 2 — Admin MFA Evidence

User: pbl-terraform-admin | Security Credentials

